

ClickHouse

ClickHouse is an open-source, columnar OLAP database management system designed for real-time analytical queries on large datasets.

Key points

- *Purpose*: fast analytics and reporting (OLAP) on **petabyte-scale** data.
- *Storage model*: columnar — stores data by **column** for high compression and efficient I/O for aggregation queries.
- *Query engine*: vectorized execution, merge tree family of table engines, heavy use of **SIMD and parallelism** for very low-latency aggregations.
- *Scalability*: supports single-node high-performance setups and distributed clusters with **sharding and replication**.
- *Consistency model*: **eventual consistency** for distributed merges; supports tunable replication and quorum settings.
- *Data ingestion*: high-throughput **inserts** (batch and streaming), supports bulk loading, Kafka integration, and HTTP/Native clients.
- *SQL dialect*: **ANSI SQL-like** with many analytical extensions (array/map types, window functions, approximate functions like `uniqExact/uniq`).
- *Indexing*: sparse primary key (**data-skipping indices**) and secondary data-skipping indexes; no traditional B-tree indexes.
- *Use cases*: **time-series** analytics, **clickstream** and **event** analytics, BI dashboards, ETL aggregation, **real-time** monitoring.
- *Integrations*: connectors for Kafka, RabbitMQ, Spark, Airflow, Grafana, Prometheus exporters and JDBC/ODBC drivers.
- *Deployment*: runs on Linux; available as self-hosted, Kubernetes operators, and managed services (various cloud providers offer **hosted ClickHouse**).
- *Licensing*: open source under the Apache 2.0 license (core); some commercial distributions/extensions exist.

To go for ClickHouse

- You need extremely fast OLAP queries over large append-heavy datasets.
- Workloads are read/aggregate heavy (GROUP BY, ORDER BY, top-N, time-windowed analytics).
- When low-latency dashboarding or real-time analytics is required.

Not to have ClickHouse

- Transactional workloads with many small updates/deletes and strict ACID requirements.
- Complex relational OLTP with lots of row-by-row operations.

Mark your feedback at the Following Contact Points

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Credits and Gratitude

- Thank all who have mentored, taught and guided. Also, always appreciate who have supported my work.